

Jonathan Violeta

Full Stack & Agentic Systems Engineer

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Summary

Full Stack and Agentic Systems Engineer with proven experience designing high-concurrency web platforms and autonomous multi-agent engineering workflows. Author of a production-grade multi-agent orchestration framework featuring isolated parallel execution, event-driven agent dispatch, and automated evaluation pipelines. Track record of technical autonomy leading full-lifecycle product engineering, from low-latency geospatial matchmaking to enterprise cloud migrations, delivering scalable and resilient software systems.

Education

Bulacan State University | Sarmiento Campus Bachelor of Science in Information Technology • College

Information Technology R374+7VV, Kaypian Rd, SJDM, Bulacan • 2020 - 2026

Immaculate Conception Polytechnic Technical Vocational Livelihood - ICT • Senior High

Information Communication Technology RX96+4XG, Marian Rd, Poblacion, Santa Maria, 3022 Bulacan • 2018 - 2020

Experience

Juanwork BGC, Taguig City, PH (WFH)

Full Stack Web Developer | Freelancer / Internship Team Leader April 2024 - March 2026

Full Stack Developer (Apr 2024 - Jan 2026) and Internship Team Leader (Jan 2026 - Mar 2026) operating with high technical autonomy across legacy maintenance and modern cloud migration.

- Led platform modernization from legacy PHP/jQuery monolith to Next.js, React, FastAPI, and PostgreSQL, re-architecting cor freelancer and client-facing modules.
- Designed relational database schemas with Drizzle ORM and integrated payment processing workflows using Paymongo and Xendit for escrow deposits and payouts.
- Engineered custom automation scripts and core backend features to optimize daily operational workflows and system performan
- Served as Internship Team Leader, guiding junior developers in Agile sprints and presenting live technical feature demonstratic to stakeholders during conferences.

Projects

Deep-* Autonomous Multi-Agent Orchestration Suite August 2026 - Present

Autonomous multi-agent software engineering framework providing structured exploration, planning, and parallel implementation workflows across modern AI coding harnesses.

- Architected an autonomous multi-agent engineering framework that orchestrates parallel AI coding agents across multiple development environments with automated file lease isolation to prevent merge collisions.
- Engineered an event-driven terminal multiplexer dispatch layer with disk-staged payload routing, eliminating command buffer truncations and reducing background orchestrator polling overhead.
- Implemented a deterministic contract validation engine that inspects worker prompt context completeness, file boundary constraints, and verification commands before task dispatch.
- Instituted multi-agent architectural deliberation panels and a dual-tier evaluation pipeline separating offline deterministic shell t suites from model-graded benchmark evals.

Limahan — Location-Aware 5v5 Matchmaking Platform June 2026 - Present

High-concurrency sports and esports matchmaking platform digitizing nationwide competitive culture with real-time vector maps, WebSockets, and escrow tournaments.

- Engineered a location-aware matchmaking engine using PostGIS spatial indexing, delivering sub-50ms map viewport searche across thousands of sports venues.
- Architected a real-time WebSocket gateway with Redis pub/sub backplane to synchronize 5v5 competitive lobbies with minima broadcast latency.
- Designed an atomic wager and escrow state machine with database row-level locking, ensuring financial consistency and automated conflict resolution during disputed matches.
- Built an interactive tournament bracket visualizer combining WebGL canvas navigation with vector export pipelines supporting to 64-team elimination ladders.

Bentahub — Centralized Multi-Branch Retail Platform June 2026 - July 2026

Unified retail management and POS platform for multi-branch store chains, centralizing transaction records, live inventory tracking, and low-stock triggers.

- Architected multi-tenant inventory management system synchronizing catalog updates and stock levels across multiple branch locations in real-time, reducing multi-store inventory drift by 94%.
- Built conflict-free customer item reservation pipeline utilizing Redis distributed locks and threshold triggers, preventing oversell and double-booking during peak shopping events.

Mindtrack — Healthcare Patient & Appointment Management System March 2026 - March 2026

Web-based patient management and appointment scheduling platform developed for Wayside Psyche Resources Center to digitize clinical operations.

- Engineered an automated appointment scheduling and patient record management system to streamline clinical intake and consultation workflows.
- Implemented role-based administrative dashboards for clinicians and staff, generating real-time attendance reporting and pati history tracking.

VetSync — Veterinary Clinic Operations & Analytics Dashboard January 2025 - February 2025

Multi-page clinic management platform providing real-time administrative oversight, patient analytics, and customer communication channels.

- Built an administrative operations dashboard featuring user management, appointment tracking, and system configuration modules.
- Integrated real-time analytics monitoring and direct client messaging to improve patient follow-up and clinic operational throughp

E-Lock — Biometric IoT Industrial LOTO Machine Isolation June 2026 - June 2026

IoT-enabled biometric Lockout/Tagout (LOTO) safety system replacing manual physical locks with automated machine isolation to ensure high-risk equipment de-energization.

- Developed FreeRTOS-based firmware on ESP32 microcontrollers integrating AS608 biometric fingerprint authentication to actu industrial shunt-trip circuit breakers for OSHA-compliant Lockout/Tagout (LOTO) protocols.
- Engineered interrupt-driven hardware isolation circuit achieving <200ms emergency power shutoff, featuring encrypted local fla audit logging to ensure tamper-proof safety compliance.

Skills

☆ **Programming Languages & Core Frameworks**

Expert

TypeScript, JavaScript, Python, PHP, C, C++, C#, HTML, CSS, TailwindCSS, NextJS, ReactJS, React Native, FastAPI, Django, Laravel, CodeIgniter, JQuery, WordPress

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☆ **Databases, Cloud & Infrastructure**

Advanced

PostgreSQL, PostGIS, MySQL, DrizzleORM, Redis, RestAPI, Docker, Vercel, Neon, Hostinger, Xampp, Wamp

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☆ **Developer Tools, Testing & Methodologies**

Advanced

Git, Github, BitBucket, Postman, VScode, Test Driven Development, Automated Testing (Playwright, Vitest), CI/CD, Agile Development, Web Security, Code Optimization



Languages

English

Upper-Intermediate level



Filipino

Native



Japanese

Beginner

